

This notebook analyzes aviation accident data to identify aircraft damage patterns and overall risk. The goal is to identify which aircraft have the lowest risk for the company to start this new company.

```
In [1]: # Import Libraries  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
In [2]: #Load the dataset  
aviation_df = pd.read_csv("AviationData.csv", encoding='latin1')
```

C:\Users\FRANKLINE\anaconda3\envs\learn-env\lib\site-packages\IPython\core\interactiveshell.py:3145: DtypeWarning: Columns (6,7,28) have mixed types. Specify dtype option on import or set low_memory=False.

```
has_raised = await self.run_ast_nodes(code_ast.body, cell_name,
```

Data understanding

In [3]: *#Showing key columns which is important for analysis of this project.*

```
aviation_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 88889 entries, 0 to 88888
Data columns (total 31 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Event.Id                             88889 non-null  object
1   Investigation.Type                    88889 non-null  object
2   Accident.Number                      88889 non-null  object
3   Event.Date                           88889 non-null  object
4   Location                             88837 non-null  object
5   Country                             88663 non-null  object
6   Latitude                             34382 non-null  object
7   Longitude                            34373 non-null  object
8   Airport.Code                         50249 non-null  object
9   Airport.Name                         52790 non-null  object
10  Injury.Severity                      87889 non-null  object
11  Aircraft.damage                      85695 non-null  object
12  Aircraft.Category                    32287 non-null  object
13  Registration.Number                  87572 non-null  object
14  Make                                 88826 non-null  object
15  Model                               88797 non-null  object
16  Amateur.Built                       88787 non-null  object
17  Number.of.Engines                   82805 non-null  float64
18  Engine.Type                         81812 non-null  object
19  FAR.Description                     32023 non-null  object
20  Schedule                            12582 non-null  object
21  Purpose.of.flight                   82697 non-null  object
22  Air.carrier                         16648 non-null  object
23  Total.Fatal.Injuries                 77488 non-null  float64
24  Total.Serious.Injuries               76379 non-null  float64
25  Total.Minor.Injuries                 76956 non-null  float64
26  Total.Uninjured                     82977 non-null  float64
27  Weather.Condition                   84397 non-null  object
28  Broad.phase.of.flight                61724 non-null  object
29  Report.Status                       82508 non-null  object
30  Publication.Date                     75118 non-null  object
dtypes: float64(5), object(26)
memory usage: 21.0+ MB
```

Data cleaning

In [4]:

```
#Keep only airplane accidents
aviation_df = aviation_df[aviation_df["Aircraft.Category"] == "Airplane"]
```

In [5]: *# Look at the first few rows*
 aviation_df.head()

Out[5]:

	Event.Id	Investigation.Type	Accident.Number	Event.Date	Location	Country	Latit
5	20170710X52551	Accident	NYC79AA106	1979-09-17	BOSTON, MA	United States	42.4
7	20020909X01562	Accident	SEA82DA022	1982-01-01	PULLMAN, WA	United States	I
8	20020909X01561	Accident	NYC82DA015	1982-01-01	EAST HANOVER, NJ	United States	I
12	20020917X02148	Accident	FTW82FRJ07	1982-01-02	HOMER, LA	United States	I
13	20020917X02134	Accident	FTW82FRA14	1982-01-02	HEARNE, TX	United States	I

5 rows × 31 columns



In [6]: *#keep only US accidents*
 aviation_df = aviation_df[aviation_df["Country"] == "United States"]

In [7]: *#Rows with missing aircraft make or model*
 aviation_df = aviation_df.dropna(subset=["Make", "Model"])

In [8]: *#Rows with missing value*
 aviation_df = aviation_df.dropna(subset=["Make", "Injury.Severity"])

In [15]: *#Handling missing values filling with unknown categorical columns*
 cat_cols = aviation_df.select_dtypes(include="object").columns
 aviation_df[cat_cols] = aviation_df[cat_cols].fillna("Unknown")

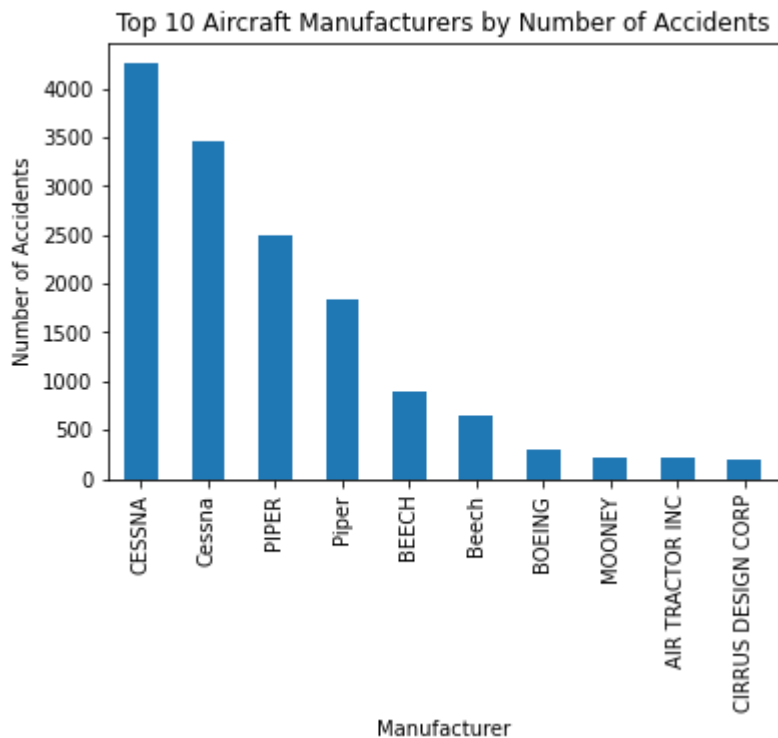
In [16]: *#Remove duplicate records*
 aviation_df = aviation_df.drop_duplicates()

In [17]: *#Save cleaned dataset*
 aviation_df.to_csv("cleaned_aviation_data.csv", index=False)

Accidents by Aircraft Manufacturer

In [25]: *#Count accidents by manufacturer*
 accidents_by_make = aviation_df["Make"].value_counts().head(10)

```
In [26]: #Plot the data
accidents_by_make.plot(kind="bar", title="Top 10 Aircraft Manufacturers by Num
plt.xlabel("Manufacturer")
plt.ylabel("Number of Accidents")
plt.show()
```



Fatal Accidents by Aircraft Model

```
In [18]: #Create a fatal accident column
aviation_df["Fatal_Accident"] = aviation_df["Total.Fatal.Injuries"] > 0
```

```
In [19]: #Calculate fatal accident rate by model
fatal_rate = aviation_df.groupby("Model")["Fatal_Accident"].mean()
```

```
In [20]: #Show lowest fatality rates
fatal_rate.sort_values().head(10)
```

```
Out[20]: Model
0-1A      0.0
MS893E    0.0
MS894A    0.0
MT 7-235  0.0
MT 7-260  0.0
MT-7-235  0.0
MT-7-260  0.0
MT7        0.0
MT7-235    0.0
MU-2       0.0
Name: Fatal_Accident, dtype: float64
```

Some aircraft models have much lower fatal accident rates, making them safer choices.

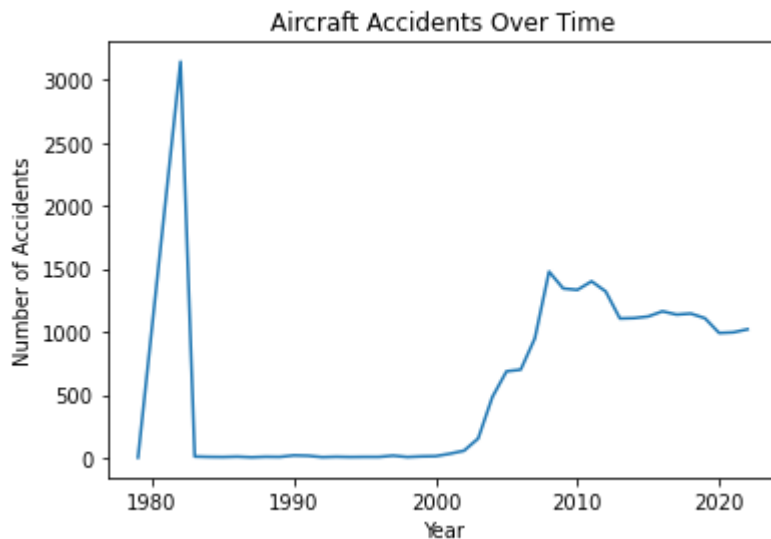
Accidents Over Time

```
In [21]: #Convert date column to datetime
aviation_df["Event.Date"] = pd.to_datetime(aviation_df["Event.Date"])
```

```
In [22]: #Extract year
aviation_df["Year"] = aviation_df["Event.Date"].dt.year
```

```
In [23]: #Count accidents per year
accidents_per_year = aviation_df.groupby("Year").size()
```

```
In [24]: #Plot trend
accidents_per_year.plot(title="Aircraft Accidents Over Time")
plt.xlabel("Year")
plt.ylabel("Number of Accidents")
plt.show()
```



It shows Aircraft safety has improved over time, especially in recent years.

Final Business Recommendations

Prioritize buying aircraft with low accident occurrence.

Opt for manufacturers with regularly reduced accidents This helps in reducing cost for maintenance and insurance.

Target on latest aircraft designs latest aircraft shows safety has improved over time